

Continuous Assessment Sheet 2 (Part B)

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Due 17:00 Wednesday 25th February 2026

1. Consider the 1D dynamical system

$$\dot{y} = (y - a)(y - b)(y - c), \quad y(t = 0) = y_0, \quad y \in \mathbb{R}, \quad (1.1)$$

where, $0 < a < b < c$ are real constants

- (a) Determine the fixed-points in terms of a , b and c .
- (b) Determine the stability of each of the fixed points.
- (c) Now set $a = b = c$. Sketch the phase-plane on the real-line in this case indicating the stability using arrows.
- (d) Again set $a = b = c$. Show that the solution blows-up at a time

$$t = \frac{1}{2(a - y_0)^2}.$$

2. We are going to consider the system

$$\ddot{x} + \dot{x} + x = 1, \quad x \in \mathbb{R}. \quad (1.2)$$

- (a) Write the system in the form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{c}, \quad (1.3)$$

where $\mathbf{x}$ and $\mathbf{c}$ are 2-dimensional vectors and $\mathbf{A}$ is a 2×2 matrix, that are to be determined.

- (b) Determine the fixed-point, $\mathbf{x}_e$.
- (c) By writing $\mathbf{x} = \mathbf{x}_e + \mathbf{v}(t)$, write the system as

$$\dot{\mathbf{v}} = \mathbf{B}\mathbf{v}, \quad (1.4)$$

where $\mathbf{B}$ is to be determined.

- (d) By solving the linear problem for $\mathbf{v}$, sketch the phase-space in the $\mathbb{R}^2$ plane.